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Locusts as Pests

The three main pest species of locusts in Australia are the Australian plague locust (Chortoicetes terminifera), the spur-throated locust (Austracris guttulosa) and the migratory locust (Locusta migratoria).

Locusts can cause widespread and severe damage to pastures, cereal crops and forage crops. They may also damage vegetable and orchard crops. The Australian plague locust is the most serious pest species in Australia due to the frequency of outbreaks (gregarious population increases) and the large areas infested. Damage to cereal crops during plagues is estimated to be over A$25 million. An economic analysis of APLC locust control during 1999-2004 concluded there was a direct benefit-cost ratio of approximately 8:1.

Dense aggregations of locust nymphs (hoppers) are called bands. Bands may extend over several kilometres and are often visible from the air. Damage by Australian plague locust hopper bands is mainly confined to pasture, and losses have been estimated to be equivalent to a 10 per cent increase in stocking rate. Hopper bands also damage cereal crops, often causing significant damage along the edges of advanced crops. In dry conditions, however, less advanced and more open crops are highly susceptible to locust infestation.

Aerial photo of damage to a wheat crop caused
by bands of Australian plague locust nymphs

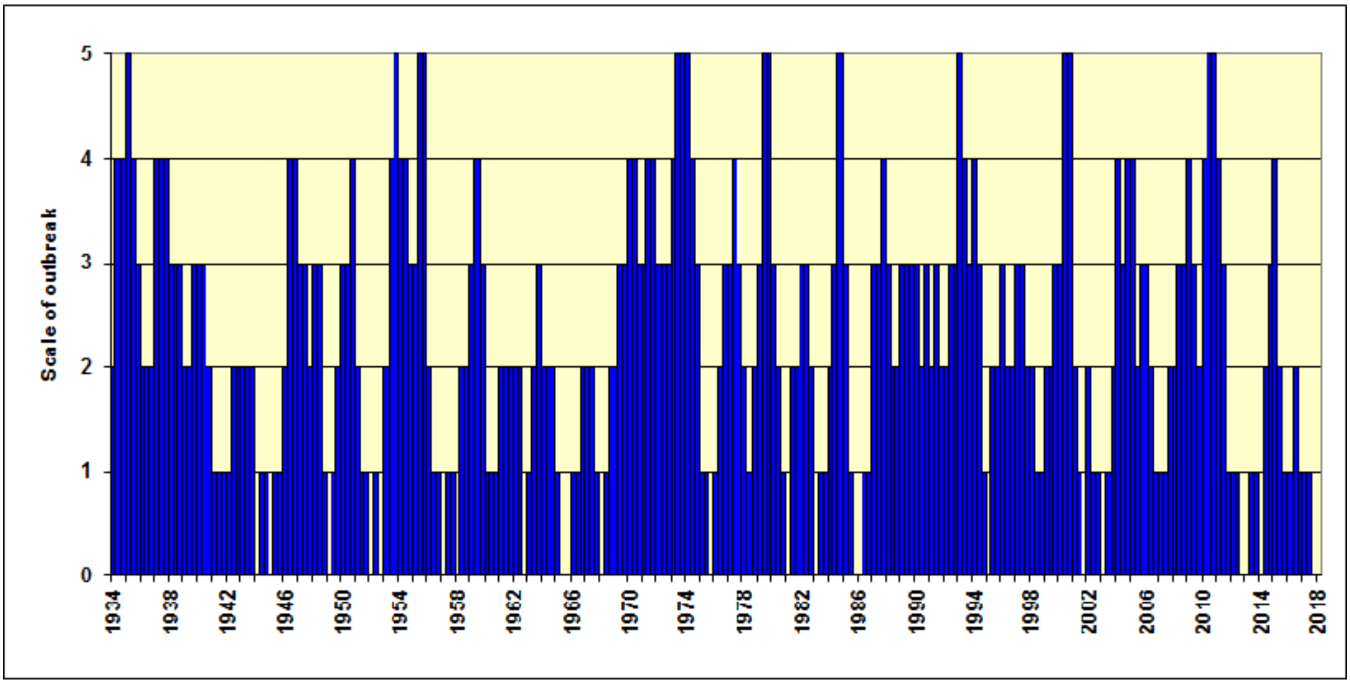
An aggregation of adult locusts is called a swarm. A swarm may contain millions of locusts and can cover an area of several square kilometres. Young winter cereals are very susceptible to damage in the autumn when locust populations usually reach their peak. Winter grain crops have usually hardened off by the time adult Australian plague locusts become active in early summer, but damage to ripening heads or contamination of harvested grain sometimes occurs.

A swarm of Locusta migratoria in central Queensland

Adults of the Australian plague locust and spur-throated locust are capable of migrating over very large distances. Overnight migrations up to several hundred kilometres are not uncommon and this behaviour can lead to the sudden appearance of large numbers of locusts in previously uninfested areas. Australian plague locust nymphs are estimated to consume 0.04 gm of green vegetation per day, averaged across all development stages. Average adult food intake has been estimated to be about 0.2 grams of green vegetation per day. With densities of adults in settled swarms ranging from about 4 to 50/m2, a swarm covering an area of 1km2 could consume over 1000 kg of green vegetation a day, depending on density.

Australian Plague Locust

Geographically extensive outbreaks were first reported in the 1870s and affected inland agricultural areas of New South Wales, South Australia and Victoria. A pattern of high density populations developing in some locations in most years, with less frequent very large 'plague' populations extending across several states for one or two years, has persisted in eastern Australia since that time. A review of locust records by the APLC identified eight major plagues since 1930. In Western Australia outbreaks were less frequent, but in recent decades major outbreaks have occurred in 1999-2001, 2004, 2006, 2009 and 2012-2013.



The years are divided into three seasons: summer, autumn and spring.

Scale of the outbreaks:

- very few locusts
- background population, a few bands/swarms
- outbreak, localised bands/swarms in one region
- major outbreaks, many bands/swarms in several regions
- plague, several hundred thousand hectares infested by bands/swarms in several regions
- major plague, over 500,000 hectares infested by bands/swarms in the agricultural zone

The scale for the graph above is taken from estimates of the area of land infested by locusts based on historical reviews by Casimir (1962), Farrow (1977), Key (1938), Magor (1970), Wright (1987) and APLC Locust Bulletins (1977-2018).

The Australian plague locust can reach plague proportions within a year if a sequence of widespread heavy rains occurs in inland areas, particularly during summer, allowing them to complete several generations of increase. Less regular rains, falling in both the interior and in the agricultural zone of eastern Australia, can maintain high density gregarious populations for several years, and continue a plague cycle. Prolonged dry periods usually result in a population decline to background levels.

An outbreak cycle may involve exchange migrations between regions of summer and winter rainfall, and the persistence of high density populations in agricultural regions of inland southeastern Australia. Heavy summer rainfalls in western Queensland often lead to large population increases and subsequent southward migrations in late summer and autumn. This pattern contributed to the development of several recorded major plagues.

Spur-throated Locust

A plague of spur-throated locusts occurred 1973-75 and caused extensive damage to crops in Queensland. Control was also conducted on outbreaks in 1994-1997, 2000-2001, mainly in cropping areas of central Queensland. A large outbreak, with widespread summer breeding in Queensland and New South Wales, occurred during 2010-11. Because the spur-throated locust has only one generation per year, it may take several years for populations of this species to develop to plague level.

Migratory Locust

This species is a major pest in many parts of the world but in Australia outbreaks have been infrequent and largely restricted to central Queensland. Early infestations are not well documented. A major outbreak occurred during the wet years between 1973 and 1976 causing severe damage to crops in Queensland and northern New South Wales. Widespread clearing of scrub and summer cropping in central Queensland greatly expanded the area of favourable habitat for this species. Since then, smaller outbreaks requiring control occurred intermittently during the 1990s, in 2000-2001 and in 2015.

Infestations of Other Species in Australia

Other species of locusts and grasshoppers that can reach outbreak numbers and occasionally cause economic damage include the small plague grasshopper (Austroicetes cruciata), wingless grasshopper (Phaulacridium vittatum), yellow winged locust (Gastrimargus musicus) and eastern plague grasshopper (Oedaleus australis). As these species have a low migratory capacity, outbreaks are usually localised and do not pose an interstate threat to agriculture. The APLC is therefore not responsible for their control. During severe infestations state authorities may assist landholders with the control of economically damaging populations.

Outbreaks of the small plague grasshopper have been recorded since 1843 and this species caused severe damage to cereal crops in South Australia during the nineteenth century. Infestations also occurred in Western Australia until the 1950s. More recent infestations occurred in South Australia in 1997 and 2006.

The wingless grasshopper is mainly a pest of improved pasture in the tablelands and western slopes of New South Wales, where average annual rainfall is above 500 mm. Outbreaks have been recorded in New South Wales since 1935 but became more severe after the introduction of improved pastures. Localised infestations of wingless grasshoppers occur frequently, causing damage to pastures, trees and gardens.

Outbreaks of the yellow-winged locust occur frequently in parts of eastern Queensland, while localised outbreaks of the eastern plague grasshopper occur irregularly. Locally high numbers of both species are often recorded in parts of Queensland and New South Wales.

Further Reading

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